

Report on the Calibration of the 2025 ^4He Beamtime

The calibration was performed using the *CaLib* tool developed by the Basel group. CaLib stores the calibration parameters in a MySQL database, which can be accessed directly by AcqRoot. Alternatively, the calibration constants can be stored in the detector setup files in the `data` directory of `acqu_user`.

The following detector calibrations were investigated:

- PID timing and ϕ angle,
- CB timing,
- CB energy,
- Tagger timing.

As a first step, I used 10 runs from the beginning of the beamtime (runset 1:CBTagg_31840–CBTagg_31849) to determine the calibration parameters. The stability of the resulting calibration was then tested by applying these parameters to a set of full-target runs taken near the end of the beamtime (runset 2: CBTagg_32330–CBTagg_32339).

1 PID Timing and ϕ -Angle Calibration

To align the PID timing, the time difference between each pair of PID hits is determined and the resulting distributions are fitted with Gaussian functions. Figure 1 shows the resulting peak positions before and after the calibration.

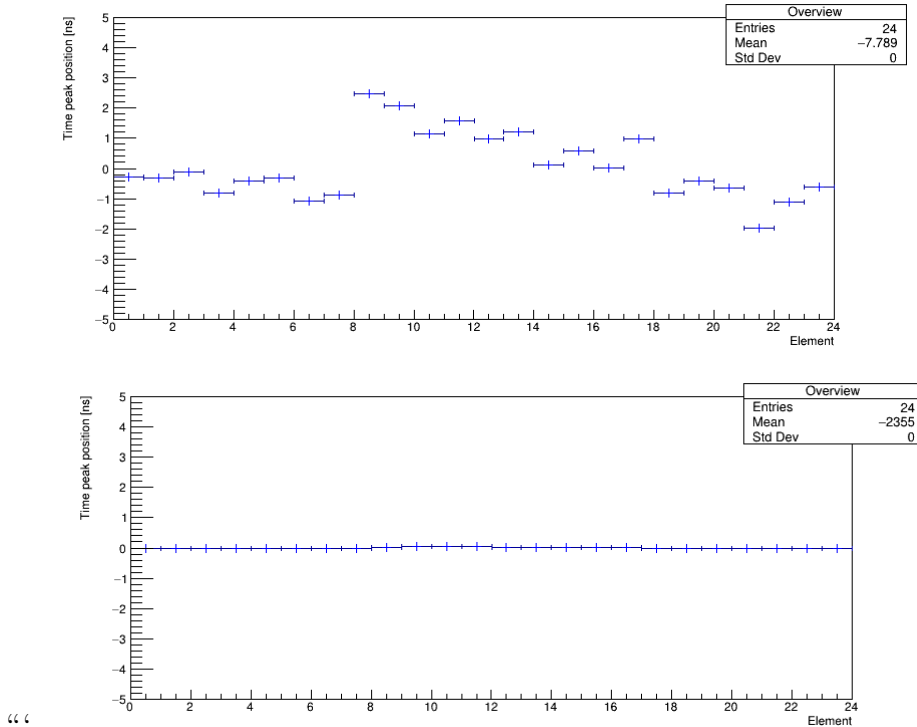


Figure 1: PID timing before (top) and after (bottom) calibration.

For the PID ϕ -angle calibration, events with exactly one hit in the CB and a coincident hit in the PID are selected. For each PID element, the ϕ angle of the corresponding CB hit is

plotted and the resulting distribution is fitted with a Gaussian function. Compared with the previously used calibration constants, the resulting PID position corresponds to a rotation of approximately 7° .

2 CB Timing

For the CB time calibration, the time differences between all photon candidates (neutral hits) in the CB are determined and the resulting distributions are fitted with Gaussian functions. From the fitted mean value, a new time offset is calculated according to

$$\text{Offset}_{\text{new}} = \text{Offset}_{\text{old}} \frac{\text{mean}}{\text{gain}}. \quad (1)$$

The gain is set to 0.11771 in the file `CaLib/config/config.cfg`. The calibration converged after approximately six iterations.

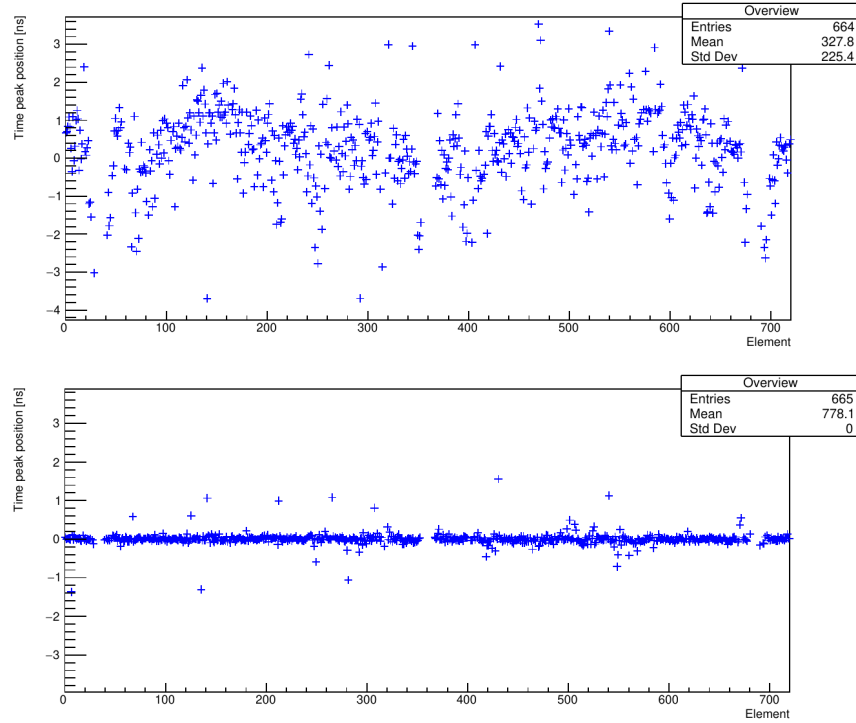


Figure 2: CB timing before (top) and after (bottom) calibration. Only data from runset 1 are used.

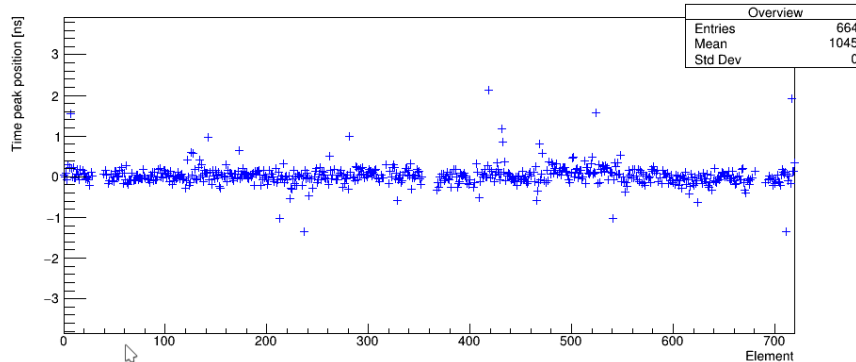


Figure 3: CB timing for runset 2 using the calibration parameters obtained from runset 1.

3 CB Energy Calibration

The aim of the CB energy calibration is to align the two-photon invariant-mass peak with the nominal π^0 mass. For this purpose, the invariant mass of two photon candidates (neutral hits) in the CB is plotted as a function of the central crystal of the corresponding cluster. The invariant-mass distribution is projected separately for each crystal and fitted with a Gaussian function.

From the fitted peak position, the new gain is calculated according to

$$\text{gain} * \text{new} = \text{gain} * \text{old} \frac{m_{\pi^0}^2}{\text{mean}^2}. \quad (2)$$

The calibration converged after approximately six iterations.

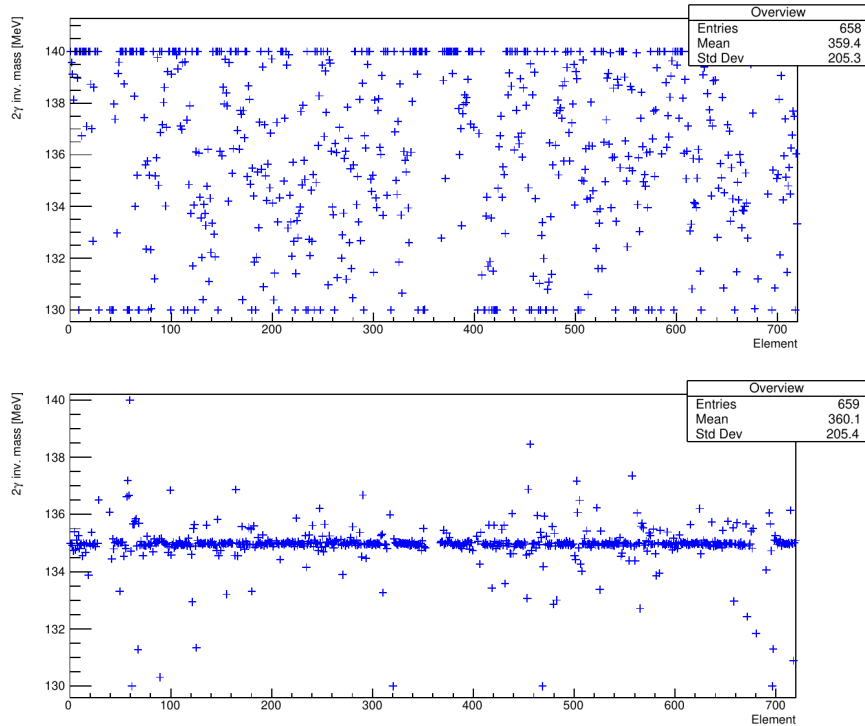


Figure 4: CB energy calibration: $\gamma\gamma$ invariant-mass peak as a function of the central crystal number before (top) and after (bottom) calibration. Only data from runset 1 are used.

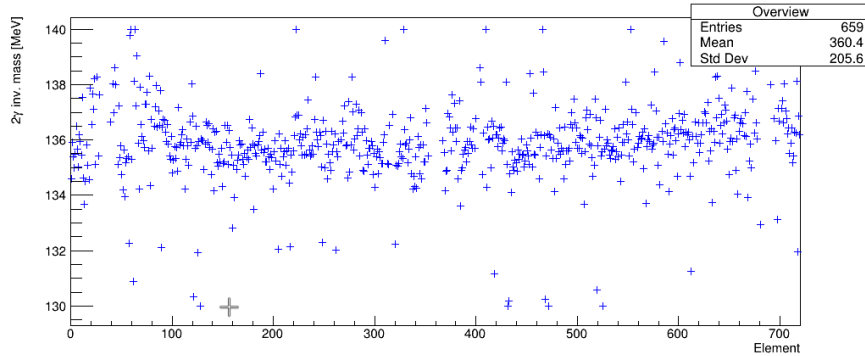


Figure 5: CB energy calibration: $\gamma\gamma$ invariant-mass peak as a function of the central crystal number for runset 2, using the calibration parameters obtained from runset 1.

Some changes in the peak positions are visible between the two runsets, indicating a variation of the CB energy response over the course of the beamtime. If a more accurate calibration

is required, the data should therefore be divided into several groups of runs, with separate calibration parameters determined for each group.

4 Tagger Timing

In the original CaLib implementation, the time difference between the Tagger and TAPS was used for the Tagger time calibration after the TAPS timing had been calibrated. In the current implementation, I instead use the TDC spectrum of each Tagger channel directly. This approach could potentially be improved by using time-difference spectra, for example between the Tagger and CB or between different Tagger channels.

For each Tagger channel, the prompt peak is fitted using a Gaussian function together with a background contribution. The calibration offsets are then adjusted such that the fitted prompt-peak positions are shifted to zero.

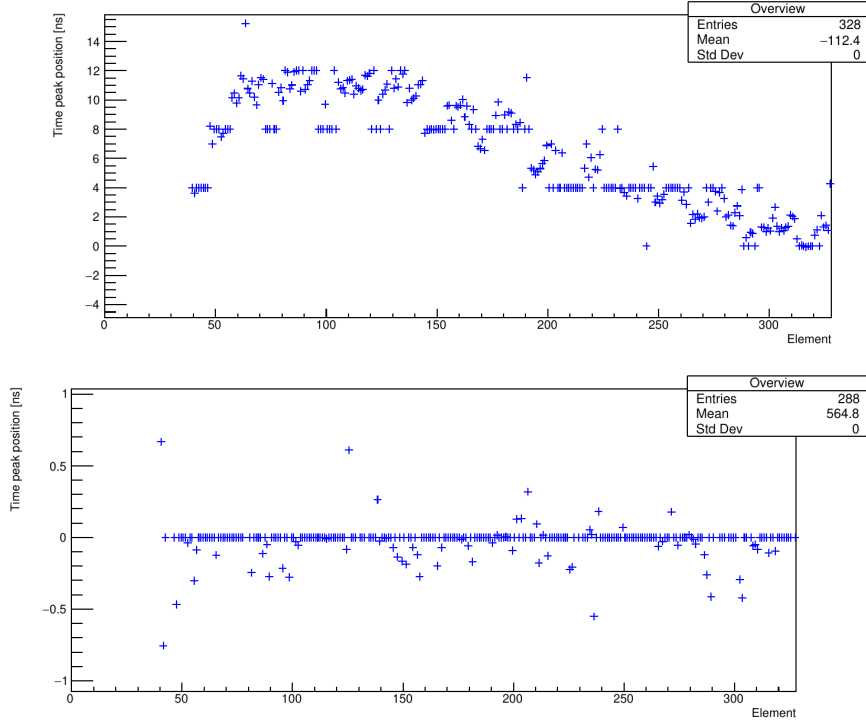


Figure 6: Position of the prompt peak in the Tagger TDC spectra before (top) and after (bottom) calibration using runset 1.

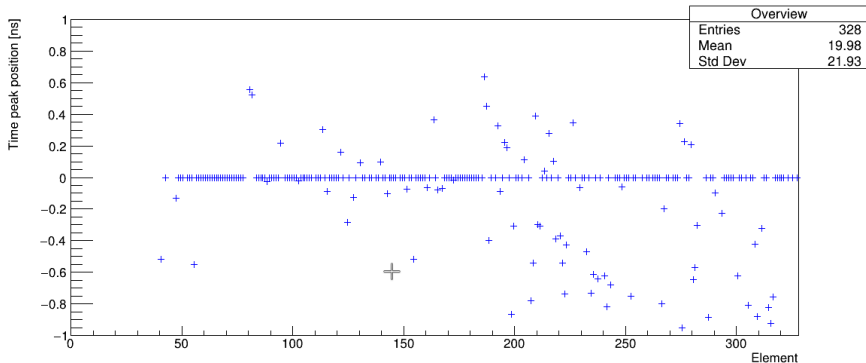


Figure 7: Position of the prompt peak in the Tagger TDC spectra for runset 2 using the calibration parameters obtained from runset 1.